

Econometrics Assignment 2

Subject code: DA242

Ramakrishna Mission Vivekananda Educational and Research Institute
M.Sc. Big Data Analytics, CS Department
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1 Question 1

In each of the following data set given in Practical Exercise 1:

- (a) Fit the linear regression of y on x . Interpret the values of the regression coefficients and calculate r^2 and comment on the goodness of fit of the linear regression.
- (b) Draw the scatter diagram and show the fitted line on it.
- (c) Also fit the reverse regression equation and hence find the r^2 and the center of gravity of the scatter diagram.

1.1 Data set 1

y_1 :	0.8	1.2	1.5	1.8	2.2	2.3	2.6	3.1
x_1 :	1.7	2.7	3.6	4.6	5.7	6.7	8.1	12.0

Table 1: Expenditure on a commodity and income of 8 persons

Here, y_1 is per capita expenditure on a commodity and x_1 is per capita income.

Solution.

(a) Fit the linear regression of y on x . Let n be the number of data points. Then $n = 8$.
Now we tabulate the required calculations involving x_1 and y_1 :

$y1$	$x1$	$(y1)^2$	$(x1)^2$	$(x1)(y1)$
0.80	1.70	0.64	2.89	1.36
1.20	2.70	1.44	7.29	3.24
1.50	3.60	2.25	12.96	5.40
1.80	4.60	3.24	21.16	8.28
2.20	5.70	4.84	32.49	12.54
2.30	6.70	5.29	44.89	15.41
2.60	8.10	6.76	65.61	21.06
3.10	12.00	9.61	144.00	37.20

Table 2: Tabulated values for Data Set 1

Now we calculate the sum of the columns for our future use:

$$\sum (y1)_i = 0.80 + 1.20 + 1.50 + 1.80 + 2.20 + 2.30 + 2.60 + 3.10 = 15.50$$

$$\sum (x1)_i = 1.70 + 2.70 + 3.60 + 4.60 + 5.70 + 6.70 + 8.10 + 12.00 = 45.10$$

$$\sum (y1)_i^2 = 0.64 + 1.44 + 2.25 + 3.24 + 4.84 + 5.29 + 6.76 + 9.61 = 34.07$$

$$\sum (x1)_i^2 = 2.89 + 7.29 + 12.96 + 21.16 + 32.49 + 44.89 + 65.61 + 144.00 = 331.29$$

$$\sum (x1)_i (y1)_i = 1.36 + 3.24 + 5.40 + 8.28 + 12.54 + 15.41 + 21.06 + 37.20 = 104.49$$

The mean of $y1$ ($\bar{y1}$) and $x1$ ($\bar{x1}$) are:

$$\bar{y1} = \frac{\sum y1_i}{n} = \frac{15.50}{8} = 1.9375$$

$$\bar{x1} = \frac{\sum x1_i}{n} = \frac{45.10}{8} = 5.6375$$

The regression equation of $y1$ on $x1$ is given by $\hat{y1} = \hat{\beta}_0 + \hat{\beta}_1(x1)$.
First, we calculate the sums of squares S_{xx} and S_{xy} :

$$S_{xx} = \sum (x1)_i^2 - n(\bar{x1})^2 = 331.2900 - 8(5.6375)^2 = 77.0387$$

$$S_{xy} = \sum (x1)_i (y1)_i - n(\bar{x1})(\bar{y1}) = 104.4900 - 8(5.6375)(1.9375) = 17.1088$$

The formula for the slope β_1 is:

$$\beta_1 = \frac{S_{xy}}{S_{xx}}$$

Hence,

$$\hat{\beta}_1 = \frac{17.1088}{77.0387} = 0.2221$$

The formula for the intercept β_0 is:

$$\beta_0 = \bar{y1} - \beta_1(\bar{x1})$$

Hence,

$$\hat{\beta}_0 = 1.9375 - (0.2221)(5.6375) = 0.6855$$

The final fitted linear regression equation is:

$$\hat{y}_1 = \hat{\beta}_0 + \hat{\beta}_1 x_1 \implies \hat{\mathbf{y}}_1 = (\mathbf{0.6855}) + (\mathbf{0.2221})\mathbf{x}_1$$

Interpretation of the regression coefficients:

$\hat{\beta}_0 = 0.6855$: The intercept represents the estimated expenditure when income is 0. $\hat{\beta}_1 = 0.2221$: The slope indicates that for each one-unit increase in income (x_1), the expenditure (y_1) is estimated to increase by 0.2221 units.

The coefficient of determination r^2 is defined by:

$$r^2 = \frac{(S_{xy})^2}{S_{xx}S_{yy}}$$

First, we calculate S_{yy} :

$$S_{yy} = \sum (y_1)_i^2 - n(\bar{y}_1)^2 = 34.0700 - 8(1.9375)^2 = 4.0388$$

Hence,

$$r^2 = \frac{(17.1088)^2}{77.0387 \times 4.0388} = 0.9408$$

Comment on the goodness of fit:

The value $r^2 = 0.9408$ means that approximately 94.08% of the total variation in expenditure can be explained by the linear relationship with income. This indicates a very strong and good fit.

(b) Draw the scatter diagram and show the fitted line on it.

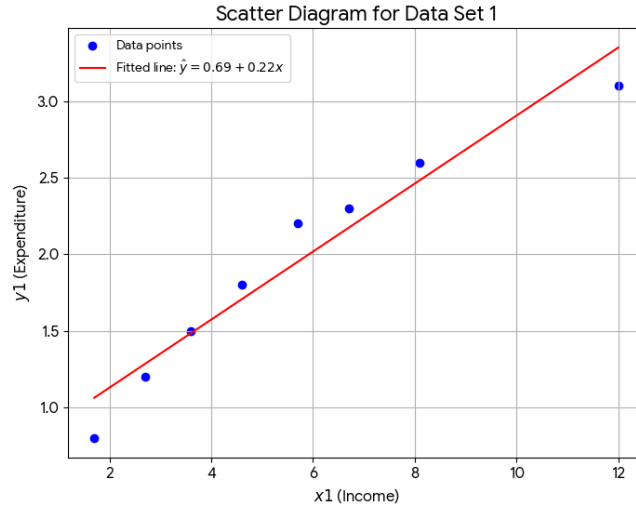


Figure 1: Scatter diagram and fitted line for Data Set 1.

(c) Fit the reverse regression equation and find the center of gravity.

The reverse regression equation of x_1 on y_1 is given by $\hat{x}_1 = \hat{\gamma}_0 + \hat{\gamma}_1 y_1$.

The formula for the slope γ_1 is:

$$\gamma_1 = \frac{S_{xy}}{S_{yy}}$$

Hence,

$$\hat{\gamma}_1 = \frac{17.1088}{4.0388} = 4.2361$$

The formula for the intercept γ_0 is:

$$\gamma_0 = \bar{x}_1 - \gamma_1(\bar{y}_1)$$

Hence,

$$\hat{\gamma}_0 = 5.6375 - (4.2361)(1.9375) = -2.5700$$

The final fitted reverse regression equation is:

$$\hat{x}_1 = \hat{\gamma}_0 + \hat{\gamma}_1 y_1 \implies \mathbf{\hat{x}_1 = (-2.5700) + (4.2361)y_1}$$

The r^2 value from this regression is also 0.9408, as it is symmetric with respect to both variables.

The center of gravity of the scatter diagram is the point of the means $(\bar{x}, \bar{y})$:

$$\mathbf{Center\ of\ Gravity = (\bar{x}_1, \bar{y}_1) = (5.6375, 1.9375)}$$

1.2 Data Set 2

y_2	70	65	90	95	110	115	120	140	155	150
x_2	80	100	120	140	160	180	200	220	240	260

Table 3: y_2 and x_2 values of 10 units in the sample

Solution.

(a) Fit the linear regression of y on x . Let n be the number of data points. Then $n = 10$.

Now we tabulate the required calculations:

y_2	x_2	$(y_2)^2$	$(x_2)^2$	$(x_2)(y_2)$
70.00	80.00	4900.00	6400.00	5600.00
65.00	100.00	4225.00	10000.00	6500.00
90.00	120.00	8100.00	14400.00	10800.00
95.00	140.00	9025.00	19600.00	13300.00
110.00	160.00	12100.00	25600.00	17600.00
115.00	180.00	13225.00	32400.00	20700.00
120.00	200.00	14400.00	40000.00	24000.00
140.00	220.00	19600.00	48400.00	30800.00
155.00	240.00	24025.00	57600.00	37200.00
150.00	260.00	22500.00	67600.00	39000.00

Table 4: Tabulated values for Data Set 2

Now we calculate the sum of the columns:

$$\begin{aligned}\sum (y_2)_i &= 70.0 + 65.0 + 90.0 + 95.0 + 110.0 + 115.0 + 120.0 + 140.0 + 155.0 + 150.0 \\ &= 1110.00\end{aligned}$$

$$\begin{aligned}\sum (x_2)_i &= 80.0 + 100.0 + 120.0 + 140.0 + 160.0 + 180.0 + 200.0 + 220.0 + 240.0 + 260.0 \\ &= 1700.00\end{aligned}$$

$$\begin{aligned}\sum (y_2)_i^2 &= 4900.00 + 4225.00 + 8100.00 + 9025.00 + 12100.00 + 13225.00 + 14400.00 \\ &\quad + 19600.00 + 24025.00 + 22500.00 \\ &= 132100.00\end{aligned}$$

$$\begin{aligned}\sum (x_2)_i^2 &= 6400.00 + 10000.00 + 14400.00 + 19600.00 + 25600.00 + 32400.00 \\ &\quad + 40000.00 + 48400.00 + 57600.00 + 67600.00 \\ &= 322000.00\end{aligned}$$

$$\begin{aligned}\sum (x_2)_i (y_2)_i &= 5600.00 + 6500.00 + 10800.00 + 13300.00 + 17600.00 + 20700.00 \\ &\quad + 24000.00 + 30800.00 + 37200.00 + 39000.00 \\ &= 205500.00\end{aligned}$$

The mean of y_2 ($\bar{y}_2$) and x_2 ($\bar{x}_2$) are:

$$\bar{y}_2 = \frac{\sum y_{2i}}{n} = \frac{1110.0000}{10} = 111.0000$$

$$\bar{x}_2 = \frac{\sum x_{2i}}{n} = \frac{1700.0000}{10} = 170.0000$$

The regression equation is given by $\hat{y}_2 = \hat{\beta}_0 + \hat{\beta}_1 x_2$.
First, we calculate S_{xx} and S_{xy} :

$$S_{xx} = \sum (x_2)_i^2 - n(\bar{x}_2)^2 = 322000.00 - 10(170.00)^2 = 33000.00$$

$$S_{xy} = \sum (x_2)_i (y_2)_i - n(\bar{x}_2)(\bar{y}_2) = 205500.00 - 10(170.00)(111.00) = 16800.00$$

The formula for the slope β_1 is:

$$\beta_1 = \frac{S_{xy}}{S_{xx}}$$

Hence,

$$\hat{\beta}_1 = \frac{16800.00}{33000.00} = 0.5091$$

The formula for the intercept β_0 is:

$$\beta_0 = \bar{y} - \beta_1(\bar{x})$$

Hence,

$$\hat{\beta}_0 = 111.00 - (0.5091)(170.00) = 24.4545$$

The final fitted linear regression equation is:

$$\hat{y}_2 = \hat{\beta}_0 + \hat{\beta}_1 x_2 \implies \mathbf{\hat{y}_2 = (24.4545) + (0.5091)x_2}$$

Interpretation of the regression coefficients:

$\hat{\beta}_0 = 24.4545$: The intercept represents the estimated value of y_2 when x_2 is 0. $\hat{\beta}_1 = 0.5091$: The slope indicates that for each one-unit increase in x_2 , the value of y_2 is estimated to increase by 0.5091 units.

The coefficient of determination r^2 is:

$$r^2 = \frac{(S_{xy})^2}{S_{xx}S_{yy}}$$

First, we calculate S_{yy} :

$$S_{yy} = \sum y_2^2 - n(\bar{y})^2 = 132100.00 - 10(111.00)^2 = 8890.00$$

Hence,

$$r^2 = \frac{(16800.00)^2}{33000.00 \times 8890.00} = 0.9621$$

Comment on the goodness of fit:

The value $r^2 = 0.9621$ means that approximately 96.21% of the total variation in y_2 can be explained by the linear relationship with x_2 . This indicates a very strong and good fit.

(b) Draw the scatter diagram and show the fitted line on it.

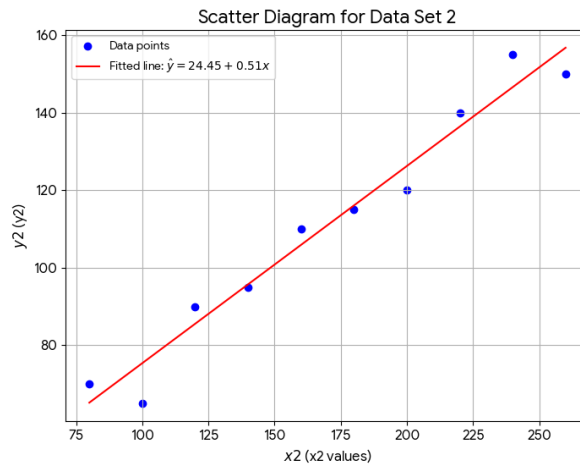


Figure 2: Scatter diagram and fitted line for Data Set 2.

(c) Fit the reverse regression equation and find the center of gravity.

The reverse regression equation is $\hat{x}_2 = \hat{\gamma}_0 + \hat{\gamma}_1 y_2$.

The formula for the slope γ_1 is:

$$\gamma_1 = \frac{S_{xy}}{S_{yy}}$$

Hence,

$$\hat{\gamma}_1 = \frac{16800.00}{8890.00} = 1.8898$$

The formula for the intercept γ_0 is:

$$\gamma_0 = \bar{x}_2 - \gamma_1(\bar{y}_2)$$

Hence,

$$\hat{\gamma}_0 = 170.0000 - (1.8898)(111.0000) = -39.7638$$

The final fitted reverse regression equation is:

$$\hat{x}_2 = \hat{\beta}_0 + \hat{\beta}_1 y_2 \implies \mathbf{\hat{x}_2} = (-\mathbf{39.7638}) + (\mathbf{1.8898})\mathbf{y_2}$$

The r^2 value is also 0.9621.

The center of gravity is $(\bar{x}, \bar{y})$:

$$\text{Center of Gravity} = (\bar{x}_2, \bar{y}_2) = (170.00, 111.00)$$

1.3 Data Set 3

y_3	256	414	634	778	1003	1839	2081	2423	2734	2914
x_3	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000

Table 5: Weekly payments to the sales representatives (y_3) and sales of the sales representatives (x_3)

Here y_3 is the weekly payment to the sales representatives based on their sales (x_3).

Solution.

(a) Fit the linear regression of y on x . Let n be the number of data points. Then $n = 10$.

Now we tabulate the required calculations:

y_3	x_3	$(y_3)^2$	$(x_3)^2$	$(x_3)(y_3)$
256.00	1000.00	65536.00	1000000.00	256000.00
414.00	2000.00	171396.00	4000000.00	828000.00
634.00	3000.00	401956.00	9000000.00	1902000.00
778.00	4000.00	605284.00	16000000.00	3112000.00
1003.00	5000.00	1006009.00	25000000.00	5015000.00
1839.00	6000.00	3381921.00	36000000.00	11034000.00
2081.00	7000.00	4330561.00	49000000.00	14567000.00
2423.00	8000.00	5870929.00	64000000.00	19384000.00
2734.00	9000.00	7474756.00	81000000.00	24606000.00
2914.00	10000.00	8491396.00	100000000.00	29140000.00

Table 6: Tabulated values for Data Set 3

Now we calculate the sum of the columns:

$$\begin{aligned}\sum (y_3)_i &= 256.00 + 414.00 + 634.00 + 778.00 + 1003.00 + 1839.00 \\ &\quad + 2081.00 + 2423.00 + 2734.00 + 2914.00 \\ &= 15076.00\end{aligned}$$

$$\begin{aligned}\sum (x_3)_i &= 1000.00 + 2000.00 + 3000.00 + 4000.00 + 5000.00 + 6000.00 \\ &\quad + 7000.00 + 8000.00 + 9000.00 + 10000.00 \\ &= 55000.00\end{aligned}$$

$$\begin{aligned}\sum (y_3)_i^2 &= 65536.00 + 171396.00 + 401956.00 + 605284.00 + 1006009.00 \\ &\quad + 3381921.00 + 4330561.00 + 5870929.00 + 7474756.00 + 8491396.00 \\ &= 31799744.00\end{aligned}$$

$$\begin{aligned}\sum (x_3)_i^2 &= 1000000.00 + 4000000.00 + 9000000.00 + 16000000.00 + 25000000.00 \\ &\quad + 36000000.00 + 49000000.00 + 64000000.00 + 81000000.00 + 100000000.00 \\ &= 385000000.00\end{aligned}$$

$$\begin{aligned}\sum (x_3)_i (y_3)_i &= 256000.00 + 828000.00 + 1902000.00 + 3112000.00 + 5015000.00 \\ &\quad + 11034000.00 + 14567000.00 + 19384000.00 + 24606000.00 + 29140000.00 \\ &= 109844000.00\end{aligned}$$

The mean of y_3 ($\bar{y}_3$) and x_3 ($\bar{x}_3$) are:

$$\bar{y}_3 = \frac{\sum (y_3)_i}{n} = \frac{15076.00}{10} = 1507.60$$

$$\bar{x}_3 = \frac{\sum (x_3)_i}{n} = \frac{55000.00}{10} = 5500.00$$

The regression equation is given by $\hat{y}_3 = \hat{\beta}_0 + \hat{\beta}_1 x_3$.

First, we calculate S_{xx} and S_{xy} :

$$S_{xx} = \sum (x_3)_i^2 - n(\bar{x}_3)^2 = 385000000.00 - 10(5500.00)^2 = 82500000.00$$

$$S_{xy} = \sum (x_3)_i (y_3)_i - n(\bar{x}_3)(\bar{y}_3) = 109844000.00 - 10(5500.00)(1507.60) = 26926000.00$$

The formula for the slope β_1 is:

$$\beta_1 = \frac{S_{xy}}{S_{xx}}$$

Hence,

$$\hat{\beta}_1 = \frac{26926000.00}{82500000.00} = 0.3264$$

The formula for the intercept β_0 is:

$$\beta_0 = \bar{y}_3 - \beta_1(\bar{x}_3)$$

Hence,

$$\hat{\beta}_0 = 1507.60 - (0.3264)(5500.00) = -287.4667$$

The final fitted linear regression equation is:

$$\hat{y}_3 = \hat{\beta}_0 + \hat{\beta}_1(x_3) \implies \mathbf{\hat{y}_3 = (-287.4667) + (0.3264)x_3}$$

Interpretation of the regression coefficients:

$\hat{\beta}_0 = -287.4667$: The intercept represents the estimated weekly payment when sales are 0. The negative value suggests the model is not applicable for zero sales. $\hat{\beta}_1 = 0.3264$: The slope indicates that for each one-unit increase in sales (x_3), the weekly payment (y_3) is estimated to increase by 0.3264 units.

The coefficient of determination r^2 is:

$$r^2 = \frac{(S_{xy})^2}{S_{xx}S_{yy}}$$

First, we calculate S_{yy} :

$$S_{yy} = \sum (y_3)_i^2 - n(\bar{y}_3)^2 = 31799744.00 - 10(1507.60)^2 = 9071166.40$$

Hence,

$$r^2 = \frac{(26926000.00)^2}{82500000.00 \times 9071166.40} = 0.9688$$

Comment on the goodness of fit:

The value $r^2 = 0.9688$ means that approximately 96.88% of the total variation in weekly payments can be explained by the linear relationship with sales. This indicates a very strong and good fit.

(b) Draw the scatter diagram and show the fitted line on it.

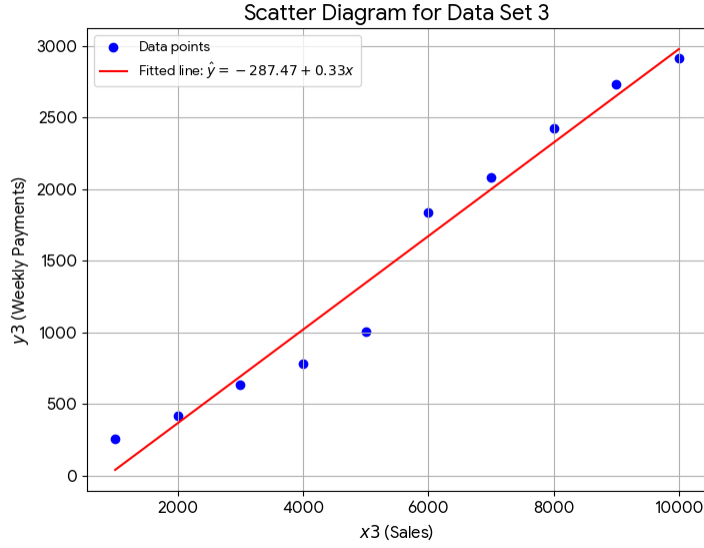


Figure 3: Scatter diagram and fitted line for Data Set 3.

(c) Fit the reverse regression equation and find the center of gravity.

The reverse regression equation is $\hat{x}_3 = \hat{\gamma}_0 + \hat{\gamma}_1 y_3$.

The formula for the slope γ_1 is:

$$\gamma_1 = \frac{S_{xy}}{S_{yy}}$$

Hence,

$$\hat{\gamma}_1 = \frac{26926000.00}{9071166.40} = 2.9683$$

The formula for the intercept γ_0 is:

$$\gamma_0 = \bar{x}_3 - \gamma_1(\bar{y}_3)$$

Hence,

$$\hat{\gamma}_0 = 5500.00 - (2.9683)(1507.60) = 1024.9815$$

The final fitted reverse regression equation is:

$$\hat{x}_3 = \hat{\gamma}_0 + \hat{\gamma}_1(y_3) \implies \mathbf{\hat{x}_3 = 1024.9815 + 2.9683y_3}$$

The r^2 value is also 0.9688.

The center of gravity is $(\bar{x}, \bar{y})$:

$$\text{Center of Gravity} = (\bar{x}_3, \bar{y}_3) = (5500.0000, 1507.6000)$$

1.4 Data Set 4

Family No.	Income (x_4)	Consumption Expenditure (y_4)
1	80	70
2	98	76
3	105	91
4	115	100
5	125	105
6	135	113
7	145	122
8	155	120
9	165	146
10	175	135
11	185	147

Table 7: Weekly consumption expenditure and income of 11 families

Solution.

(a) Fit the linear regression of y on x . Let n be the number of data points. Then $n = 11$.
Now we tabulate the required calculations:

y_4	x_4	y_4^2	x_4^2	$x_4 y_4$
70.00	80.00	4900.00	6400.00	5600.00
76.00	98.00	5776.00	9604.00	7448.00
91.00	105.00	8281.00	11025.00	9555.00
100.00	115.00	10000.00	13225.00	11500.00
105.00	125.00	11025.00	15625.00	13125.00
113.00	135.00	12769.00	18225.00	15255.00
122.00	145.00	14884.00	21025.00	17690.00
120.00	155.00	14400.00	24025.00	18600.00
146.00	165.00	21316.00	27225.00	24090.00
135.00	175.00	18225.00	30625.00	23625.00
147.00	185.00	21609.00	34225.00	27195.00

Table 8: Tabulated values for Data Set 4

Now we calculate the sum of the columns:

$$\begin{aligned}\sum (y4)_i &= 70.00 + 76.00 + 91.00 + 100.00 + 105.00 + 113.00 \\ &\quad + 122.00 + 120.00 + 146.00 + 135.00 + 147.00 \\ &= 1225.00\end{aligned}$$

$$\begin{aligned}\sum (x4)_i &= 80.00 + 98.00 + 105.00 + 115.00 + 125.00 + 135.00 \\ &\quad + 145.00 + 155.00 + 165.00 + 175.00 + 185.00 \\ &= 1483.00\end{aligned}$$

$$\begin{aligned}\sum (y4)_i^2 &= 4900.00 + 5776.00 + 8281.00 + 10000.00 + 11025.00 + 12769.00 \\ &\quad + 14884.00 + 14400.00 + 21316.00 + 18225.00 + 21609.00 \\ &= 143185.00\end{aligned}$$

$$\begin{aligned}\sum (x4)_i^2 &= 6400.00 + 9604.00 + 11025.00 + 13225.00 + 15625.00 + 18225.00 \\ &\quad + 21025.00 + 24025.00 + 27225.00 + 30625.00 + 34225.00 \\ &= 211229.00\end{aligned}$$

$$\begin{aligned}\sum (x4)_i (y4)_i &= 5600.00 + 7448.00 + 9555.00 + 11500.00 + 13125.00 + 15255.00 \\ &\quad + 17690.00 + 18600.00 + 24090.00 + 23625.00 + 27195.00 \\ &= 173683.00\end{aligned}$$

The mean of $y4$ ($\bar{y4}$) and $x4$ ($\bar{x4}$) are:

$$\begin{aligned}\bar{y4} &= \frac{\sum (y4)_i}{n} = \frac{1225.00}{11} = 111.3636 \\ \bar{x4} &= \frac{\sum (x4)_i}{n} = \frac{1483.00}{11} = 134.8182\end{aligned}$$

The regression equation is given by $\hat{y4} = \hat{\beta}_0 + \hat{\beta}_1(x4)$.
First, we calculate S_{xx} and S_{xy} :

$$S_{xx} = \sum (x4)_i^2 - n(\bar{x4})^2 = 211229.00 - 11(134.8182)^2 = 11293.6364$$

$$S_{xy} = \sum (x4)_i (y4)_i - n(\bar{x4})(\bar{y4}) = 173683.00 - 11(134.8182)(111.3636) = 8530.7273$$

The formula for the slope β_1 is:

$$\beta_1 = \frac{S_{xy}}{S_{xx}}$$

Hence,

$$\hat{\beta}_1 = \frac{8530.7273}{11293.6364} = 0.7554$$

The formula for the intercept β_0 is:

$$\beta_0 = (\bar{y}_4) - \beta_1(\bar{x}_4)$$

Hence,

$$\hat{\beta}_0 = 111.3636 - (0.7554)(134.8182) = 9.5278$$

The final fitted linear regression equation is:

$$\hat{y}_4 = \hat{\beta}_0 + \hat{\beta}_1(x_4) \implies \hat{\mathbf{y}}_4 = (\mathbf{9.5278}) + (\mathbf{0.7554})\mathbf{x}_4$$

Interpretation of the regression coefficients:

$\hat{\beta}_0 = 9.5278$: The intercept represents the estimated consumption expenditure when income is 0. $\hat{\beta}_1 = 0.7554$: The slope indicates that for each one-unit increase in income (x_4), the consumption expenditure (y_4) is estimated to increase by 0.7554 units.

The coefficient of determination r^2 is:

$$r^2 = \frac{(S_{xy})^2}{S_{xx}S_{yy}}$$

First, we calculate S_{yy} :

$$S_{yy} = \sum (y_4)_i^2 - n(\bar{y}_4)^2 = 143185.0000 - 11(111.3636)^2 = 6764.5455$$

Hence,

$$r^2 = \frac{(8530.7273)^2}{11293.6364 \times 6764.5455} = 0.9526$$

Comment on the goodness of fit:

The value $r^2 = 0.9526$ means that approximately 95.26% of the total variation in consumption expenditure can be explained by the linear relationship with income. This indicates a very strong and good fit.

(b) Draw the scatter diagram and show the fitted line on it.

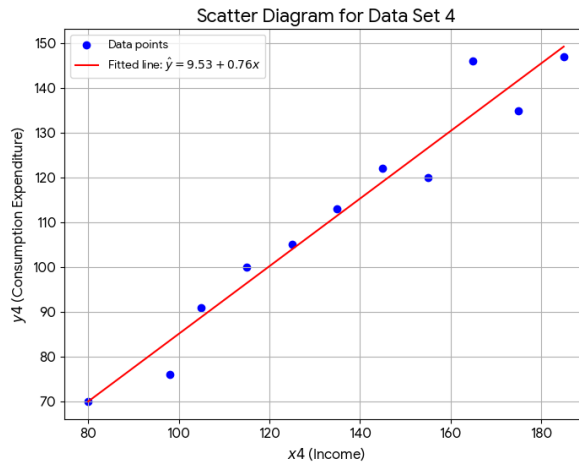


Figure 4: Scatter diagram and fitted line for Data Set 4.

(c) Fit the reverse regression equation and find the center of gravity.

The reverse regression equation is $\hat{x}_4 = \hat{\gamma}_0 + \hat{\gamma}_1(y_4)$.

The formula for the slope γ_1 is:

$$\gamma_1 = \frac{S_{xy}}{S_{yy}}$$

Hence,

$$\hat{\gamma}_1 = \frac{8530.7273}{6764.5455} = 1.2611$$

The formula for the intercept γ_0 is:

$$\gamma_0 = \bar{x}_4 - \gamma_1 \bar{y}_4$$

Hence,

$$\hat{\gamma}_0 = 134.8182 - (1.2611)(111.3636) = -5.6218$$

The final fitted reverse regression equation is:

$$\hat{x}_4 = \hat{\gamma}_0 + \hat{\gamma}_1(y_4) \implies \mathbf{\hat{x}_4 = (-5.6218) + (1.2611)y_4}$$

The r^2 value is also 0.9526.

The center of gravity is $(\bar{x}, \bar{y})$:

$$\text{Center of Gravity} = (\bar{x}_4, \bar{y}_4) = (134.8182, 111.3636)$$

1.5 Data Set 5

Marks (y_5) \ Attendance (x_5)	60%	70%	80%	90%	100%
60%	1	2	1	0	0
70%	1	3	3	1	0
80%	0	1	3	1	0
90%	0	0	4	4	0
100%	0	0	4	7	4

Table 9: Bi-variate distribution of attendance and marks in an examination for 40 students

Each cell in the table represents the number of students.

Solution.

(a) Fit the linear regression of y on x. Let n be the number of data points. Then $n = 40$.

Now we tabulate the required calculations for all 40 students:

y_5	x_5	y_5^2	x_5^2	$x_5 y_5$
50.00	60.00	2500.00	3600.00	3000.00
50.00	70.00	2500.00	4900.00	3500.00
50.00	70.00	2500.00	4900.00	3500.00

$(y5)$	$(x5)$	$(y5)^2$	$(x5)^2$	$(x5)(y5)$
50.00	80.00	2500.00	6400.00	4000.00
60.00	60.00	3600.00	3600.00	3600.00
60.00	70.00	3600.00	4900.00	4200.00
60.00	70.00	3600.00	4900.00	4200.00
60.00	70.00	3600.00	4900.00	4200.00
60.00	80.00	3600.00	6400.00	4800.00
60.00	80.00	3600.00	6400.00	4800.00
60.00	80.00	3600.00	6400.00	4800.00
60.00	90.00	3600.00	8100.00	5400.00
70.00	70.00	4900.00	4900.00	4900.00
70.00	80.00	4900.00	6400.00	5600.00
70.00	80.00	4900.00	6400.00	5600.00
70.00	80.00	4900.00	6400.00	5600.00
70.00	90.00	4900.00	8100.00	6300.00
80.00	80.00	6400.00	6400.00	6400.00
80.00	80.00	6400.00	6400.00	6400.00
80.00	80.00	6400.00	6400.00	6400.00
80.00	80.00	6400.00	6400.00	6400.00
80.00	90.00	6400.00	8100.00	7200.00
80.00	90.00	6400.00	8100.00	7200.00
80.00	90.00	6400.00	8100.00	7200.00
80.00	90.00	6400.00	8100.00	7200.00
90.00	80.00	8100.00	6400.00	7200.00
90.00	80.00	8100.00	6400.00	7200.00
90.00	80.00	8100.00	6400.00	7200.00
90.00	80.00	8100.00	6400.00	7200.00
90.00	90.00	8100.00	8100.00	8100.00
90.00	90.00	8100.00	8100.00	8100.00
90.00	90.00	8100.00	8100.00	8100.00
90.00	90.00	8100.00	8100.00	8100.00
90.00	90.00	8100.00	8100.00	8100.00
90.00	90.00	8100.00	8100.00	8100.00
90.00	90.00	8100.00	8100.00	8100.00
90.00	90.00	8100.00	8100.00	8100.00
90.00	100.00	8100.00	10000.00	9000.00
90.00	100.00	8100.00	10000.00	9000.00
90.00	100.00	8100.00	10000.00	9000.00
90.00	100.00	8100.00	10000.00	9000.00

Table 10: Tabulated values for Data Set 5

Now we calculate the sum of the columns:

$$\begin{aligned}
\sum (y5)_i &= 50.00 + 50.00 + 50.00 + 50.00 + 60.00 + 60.00 \\
&\quad + 60.00 + 60.00 + 60.00 + 60.00 + 60.00 + 60.00 \\
&\quad + 70.00 + 70.00 + 70.00 + 70.00 + 70.00 + 80.00 \\
&\quad + 80.00 + 80.00 + 80.00 + 80.00 + 80.00 + 80.00 \\
&\quad + 80.00 + 90.00 + 90.00 + 90.00 + 90.00 + 90.00 \\
&\quad + 90.00 + 90.00 + 90.00 + 90.00 + 90.00 + 90.00 \\
&\quad + 90.00 + 90.00 + 90.00 + 90.00 \\
&= 3020.00
\end{aligned}$$

$$\begin{aligned}
\sum (x5)_i &= 60.00 + 70.00 + 70.00 + 80.00 + 60.00 + 70.00 \\
&\quad + 70.00 + 70.00 + 80.00 + 80.00 + 80.00 + 90.00 \\
&\quad + 70.00 + 80.00 + 80.00 + 80.00 + 90.00 + 80.00 \\
&\quad + 80.00 + 80.00 + 80.00 + 90.00 + 90.00 + 90.00 \\
&\quad + 90.00 + 80.00 + 80.00 + 80.00 + 80.00 + 90.00 \\
&\quad + 90.00 + 90.00 + 90.00 + 90.00 + 90.00 + 90.00 \\
&\quad + 100.00 + 100.00 + 100.00 + 100.00 \\
&= 3310.00
\end{aligned}$$

$$\begin{aligned}
\sum (y5)_i^2 &= 2500.00 + 2500.00 + 2500.00 + 2500.00 + 3600.00 + 3600.00 \\
&\quad + 3600.00 + 3600.00 + 3600.00 + 3600.00 + 3600.00 + 3600.00 \\
&\quad + 4900.00 + 4900.00 + 4900.00 + 4900.00 + 4900.00 + 6400.00 \\
&\quad + 6400.00 + 6400.00 + 6400.00 + 6400.00 + 6400.00 + 6400.00 \\
&\quad + 6400.00 + 8100.00 + 8100.00 + 8100.00 + 8100.00 + 8100.00 \\
&\quad + 8100.00 + 8100.00 + 8100.00 + 8100.00 + 8100.00 + 8100.00 \\
&\quad + 8100.00 + 8100.00 + 8100.00 + 8100.00 \\
&= 236000.00
\end{aligned}$$

$$\begin{aligned}
\sum (x5)_i^2 &= 3600.00 + 4900.00 + 4900.00 + 6400.00 + 3600.00 + 4900.00 \\
&\quad + 4900.00 + 4900.00 + 6400.00 + 6400.00 + 6400.00 + 8100.00 \\
&\quad + 4900.00 + 6400.00 + 6400.00 + 6400.00 + 8100.00 + 6400.00 \\
&\quad + 6400.00 + 6400.00 + 6400.00 + 8100.00 + 8100.00 + 8100.00 \\
&\quad + 8100.00 + 6400.00 + 6400.00 + 6400.00 + 6400.00 + 8100.00 \\
&\quad + 8100.00 + 8100.00 + 8100.00 + 8100.00 + 8100.00 + 8100.00 \\
&\quad + 10000.00 + 10000.00 + 10000.00 + 10000.00 \\
&= 277900.00
\end{aligned}$$

$$\begin{aligned}
\sum (x5)_i (y5)_i &= 3000.00 + 3500.00 + 3500.00 + 4000.00 + 3600.00 + 4200.00 \\
&\quad + 4200.00 + 4200.00 + 4800.00 + 4800.00 + 4800.00 + 5400.00 \\
&\quad + 4900.00 + 5600.00 + 5600.00 + 5600.00 + 6300.00 + 6400.00 \\
&\quad + 6400.00 + 6400.00 + 6400.00 + 7200.00 + 7200.00 + 7200.00 \\
&\quad + 7200.00 + 7200.00 + 7200.00 + 7200.00 + 7200.00 + 8100.00 \\
&\quad + 8100.00 + 8100.00 + 8100.00 + 8100.00 + 8100.00 + 8100.00 \\
&\quad + 9000.00 + 9000.00 + 9000.00 + 9000.00 \\
&= 253900.00
\end{aligned}$$

The mean of $y5$ ($\bar{y5}$) and $x5$ ($\bar{x5}$) are:

$$\begin{aligned}
\bar{y5} &= \frac{\sum (y5)_i}{n} = \frac{3020.00}{40} = 75.50 \\
\bar{x5} &= \frac{\sum (x5)_i}{n} = \frac{3310.00}{40} = 82.75
\end{aligned}$$

The regression equation is given by $\hat{y5} = \hat{\beta}_0 + \hat{\beta}_1(x5)$.
First, we calculate S_{xx} and S_{xy} :

$$S_{xx} = \sum (x5)_i^2 - n(\bar{x5})^2 = 277900.00 - 40(82.75)^2 = 3997.50$$

$$S_{xy} = \sum (x5)_i (y5)_i - n(\bar{x5})(\bar{y5}) = 253900.00 - 40(82.75)(75.50) = 3995.00$$

The formula for the slope β_1 is:

$$\beta_1 = \frac{S_{xy}}{S_{xx}}$$

Hence,

$$\hat{\beta}_1 = \frac{3995.00}{3997.50} = 0.9994$$

The formula for the intercept β_0 is:

$$\beta_0 = \bar{y5} - \beta_1 \bar{x5}$$

Hence,

$$\hat{\beta}_0 = 75.50 - (0.9994)(82.75) = -7.1982$$

The final fitted linear regression equation is:

$$\hat{y}_5 = \hat{\beta}_0 + \hat{\beta}_1(x_5) \implies \hat{\mathbf{y}}_5 = (-7.1982) + (0.9994)\mathbf{x}_5$$

Interpretation of the regression coefficients:

$\hat{\beta}_0 = -7.1982$: The intercept represents the estimated marks when attendance is 0. The negative value suggests the model may not be applicable for zero attendance. $\hat{\beta}_1 = 0.9994$: The slope indicates that for each 1% increase in attendance, the marks are estimated to increase by 0.9994 points.

The coefficient of determination r^2 is:

$$r^2 = \frac{(S_{xy})^2}{S_{xx}S_{yy}}$$

First, we calculate S_{yy} :

$$S_{yy} = \sum (y_5)_i^2 - n(\bar{y}_5)^2 = 236000.00 - 40(75.50)^2 = 7990.00$$

Hence,

$$r^2 = \frac{(3995.00)^2}{3997.50 \times 7990.00} = 0.4997$$

Comment on the goodness of fit:

The value $r^2 = 0.4997$ means that approximately 49.97% of the total variation in marks can be explained by the linear relationship with attendance. This indicates a moderate fit.

(b) Draw the scatter diagram and show the fitted line on it. (given in next page)

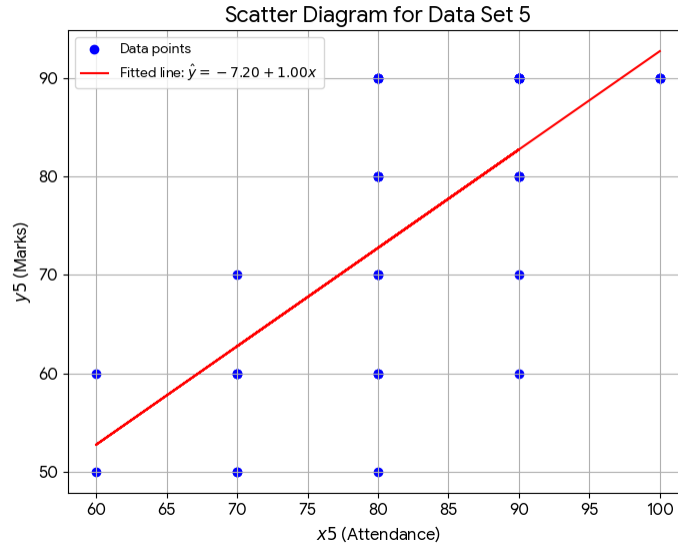


Figure 5: Scatter diagram and fitted line for Data Set 5.

(c) Fit the reverse regression equation and find the center of gravity.

The reverse regression equation is $\hat{x}\bar{5} = \hat{\gamma}_0 + \hat{\gamma}_1(y\bar{5})$. The formula for the slope γ_1 is:

$$\gamma_1 = \frac{S_{xy}}{S_{yy}}$$

Hence,

$$\hat{\gamma}_1 = \frac{3995.00}{7990.00} = 0.50$$

The formula for the intercept γ_0 is:

$$\gamma_0 = x\bar{5} - \gamma_1 y\bar{5}$$

Hence,

$$\hat{\gamma}_0 = 82.75 - (0.50)(75.50) = 45.00$$

The final fitted reverse regression equation is:

$$\hat{x}\bar{5} = \hat{\gamma}_0 + \hat{\gamma}_1(y\bar{5}) \implies \mathbf{\hat{x}\bar{5} = (45.00) + (0.50)y\bar{5}}$$

The r^2 value is also 0.4997.

The center of gravity is $(x\bar{5}, y\bar{5})$:

$$\text{Center of Gravity} = (x\bar{5}, y\bar{5}) = (82.75, 75.50)$$